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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
International General Certificate of Secondary Education
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER | | | | | |

CANDIDATE NUMBER | | | | |

2760179610

ENVIRONMENTAL MANAGEMENT

0680/04
5014/02

Alternative to Coursework

May/June 2009

1 hour 30 minutes

Candidates answer on the Question Paper.

Additional Materials: Ruler

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.
Study the appropriate Source materials before you start to write your answers.
Credit will be given for appropriate selection and use of data in your answers and for relevant interpretation of these data. Suggestions for data sources are given in some questions.
You may use the source data to draw diagrams and graphs or to do calculations to illustrate your answers.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use

This document consists of **18** printed pages and **2** blank pages.

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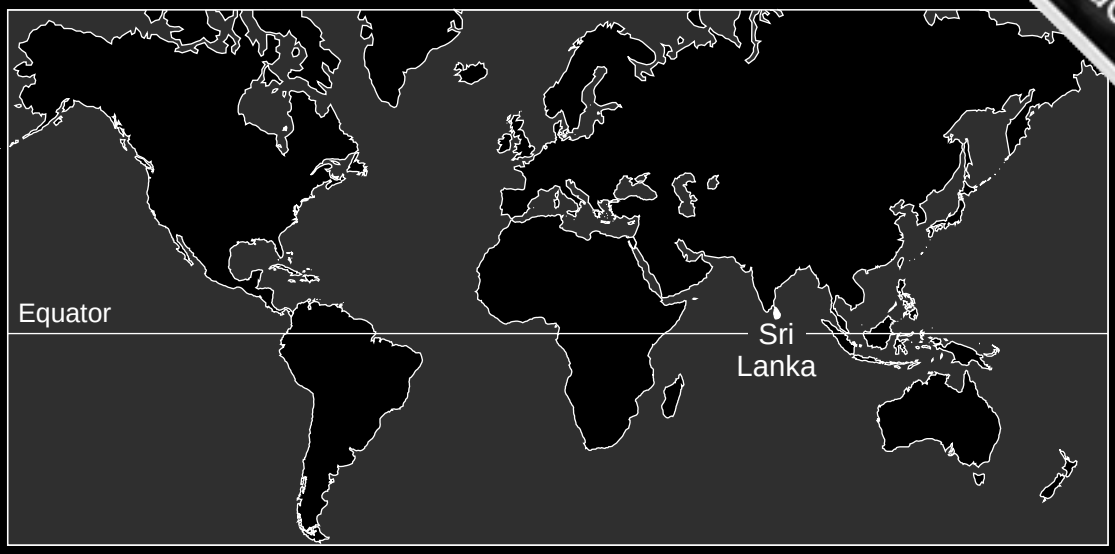


Fig. 1 map of the World

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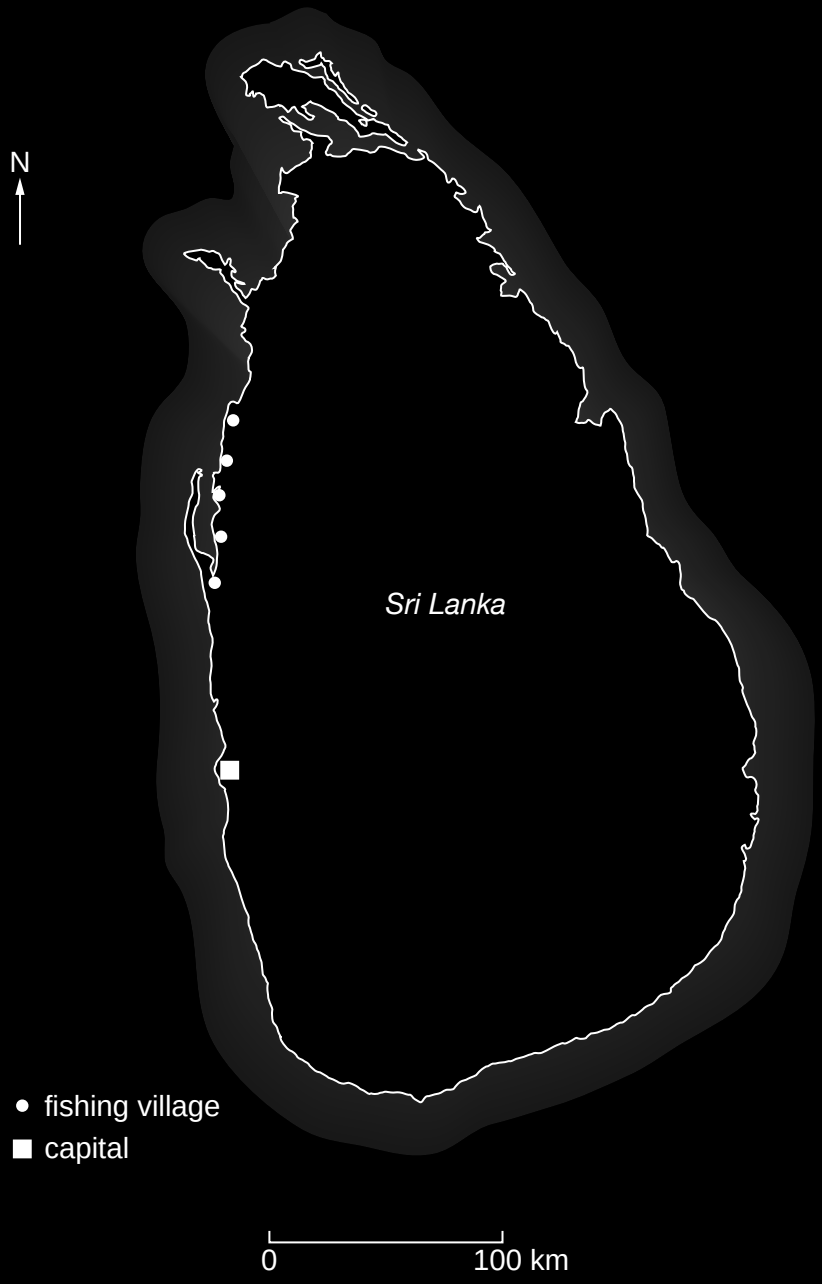


Fig. 2 map of Sri Lanka

- Area: 64 700 sq km
- Population: 21 000 000
- Children per woman: 2.05
- Life expectancy at birth: 74.8 years
- Currency: Rupee (44 Rs = 1 US Dollar)
- Official languages: Sinhala, Tamil, 14 other languages
- Climate: Tropical
- Terrain: mostly plains with mountains in the south central interior
- Main exports: textiles, tea, spices, diamonds, emeralds, rubies, fish, coconut and rubber products

Sri Lanka's main economic activities are in food processing, telecommunication, insurance and banking. Plantation crops used to be the major export but now only make up 15% of all exports. About 800 000 people work abroad, sending more than a billion dollars back to the Sri Lankan economy.

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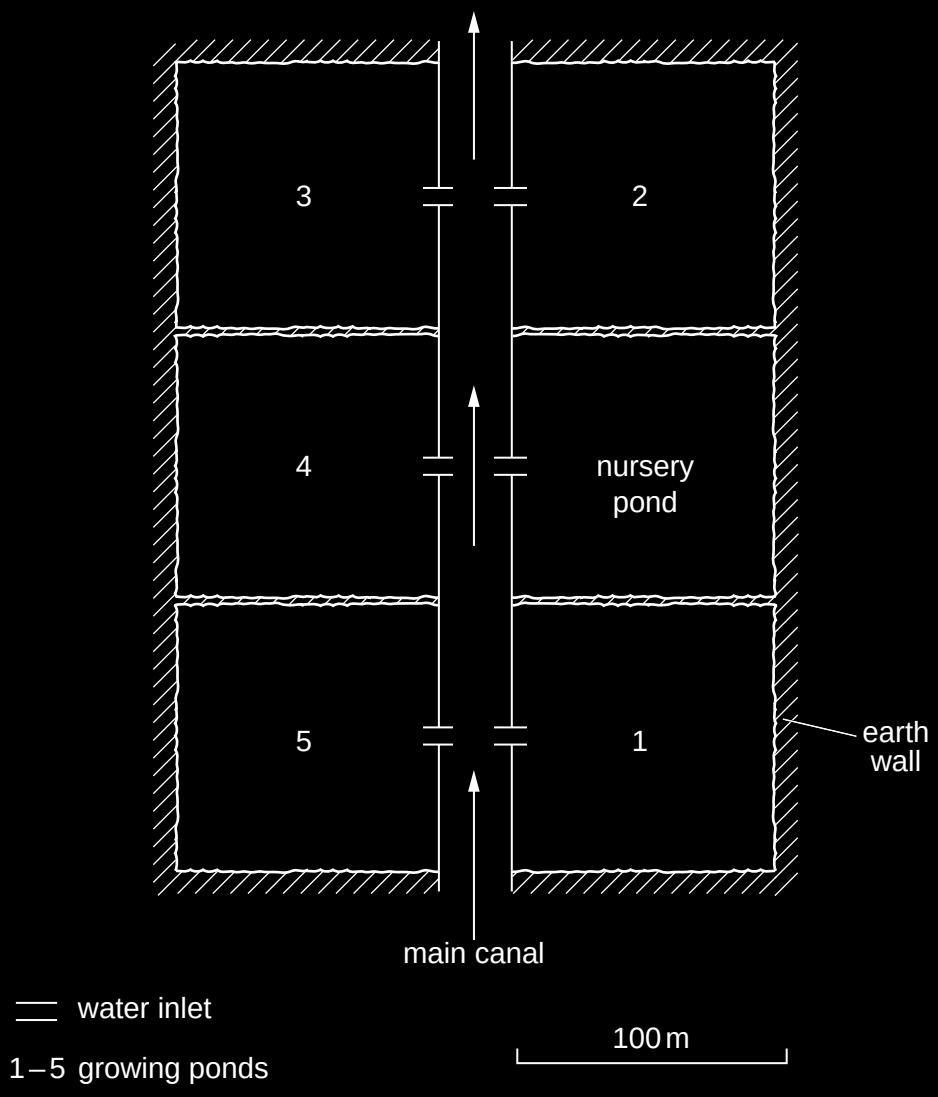


Fig. 3

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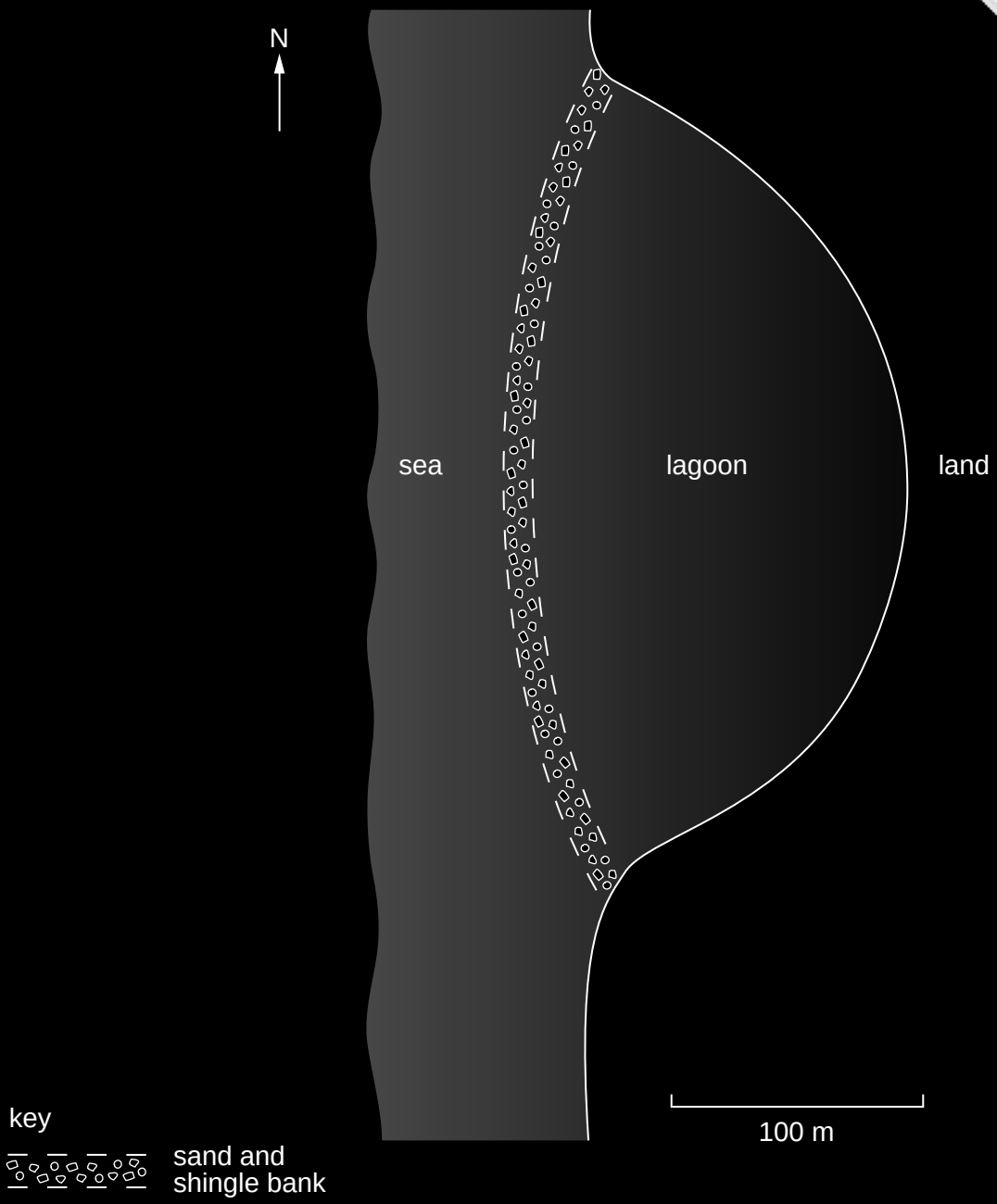


Fig. 4

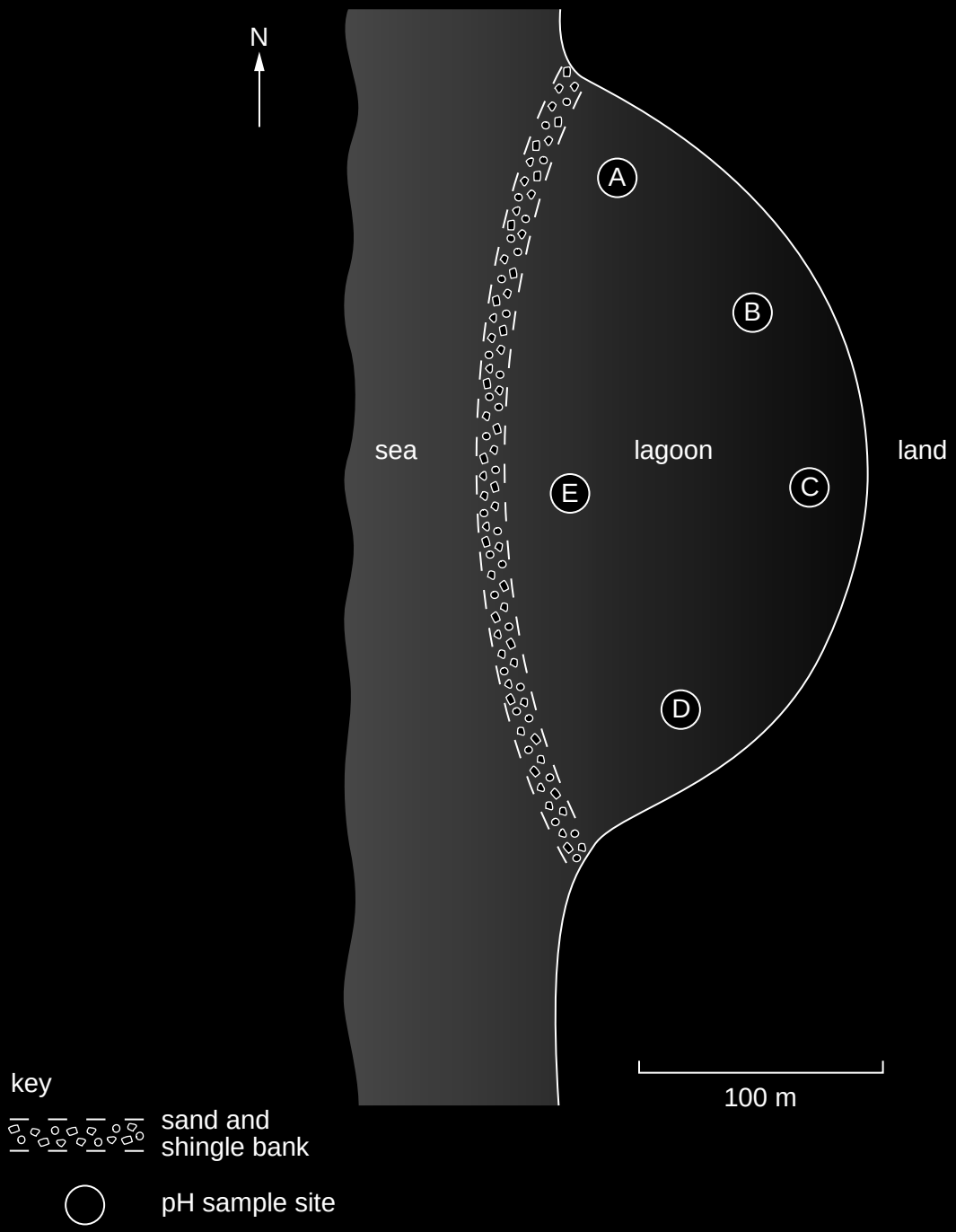
(ii) The building costs have been estimated at Rs 200 000. The shrimp are worth Rs 80 per kilogram.

How many kilograms of shrimp must be produced to pay back the building costs? (Show your working).

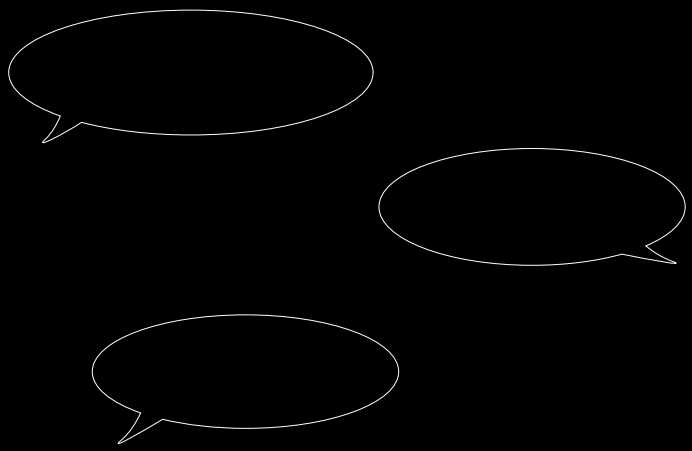
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Roof rain water harvesting system–very low cost model

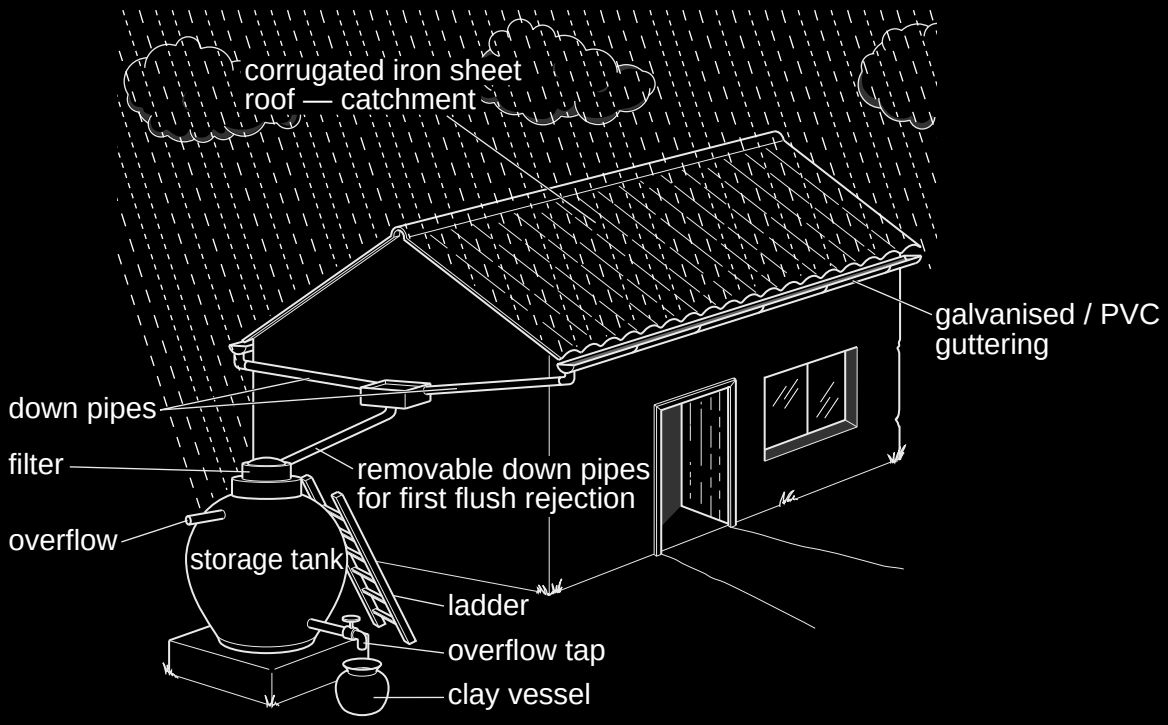


Fig. 7

(a) **(i)** This system has removable down pipes to reject the first flush of rainwater. Explain why.

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..... [2]

(ii) A fine filter is used. It stops mosquitoes entering the storage tank.
Explain how this helps keep the people healthy.

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(iii) Give another reason why a fine filter is used.

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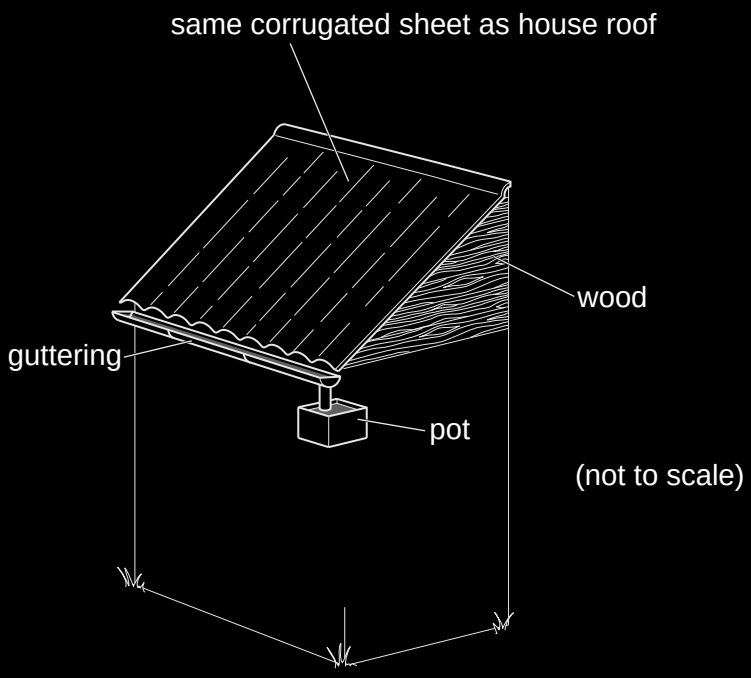


Fig. 8

(i) Why did the students use more than one rain collector?

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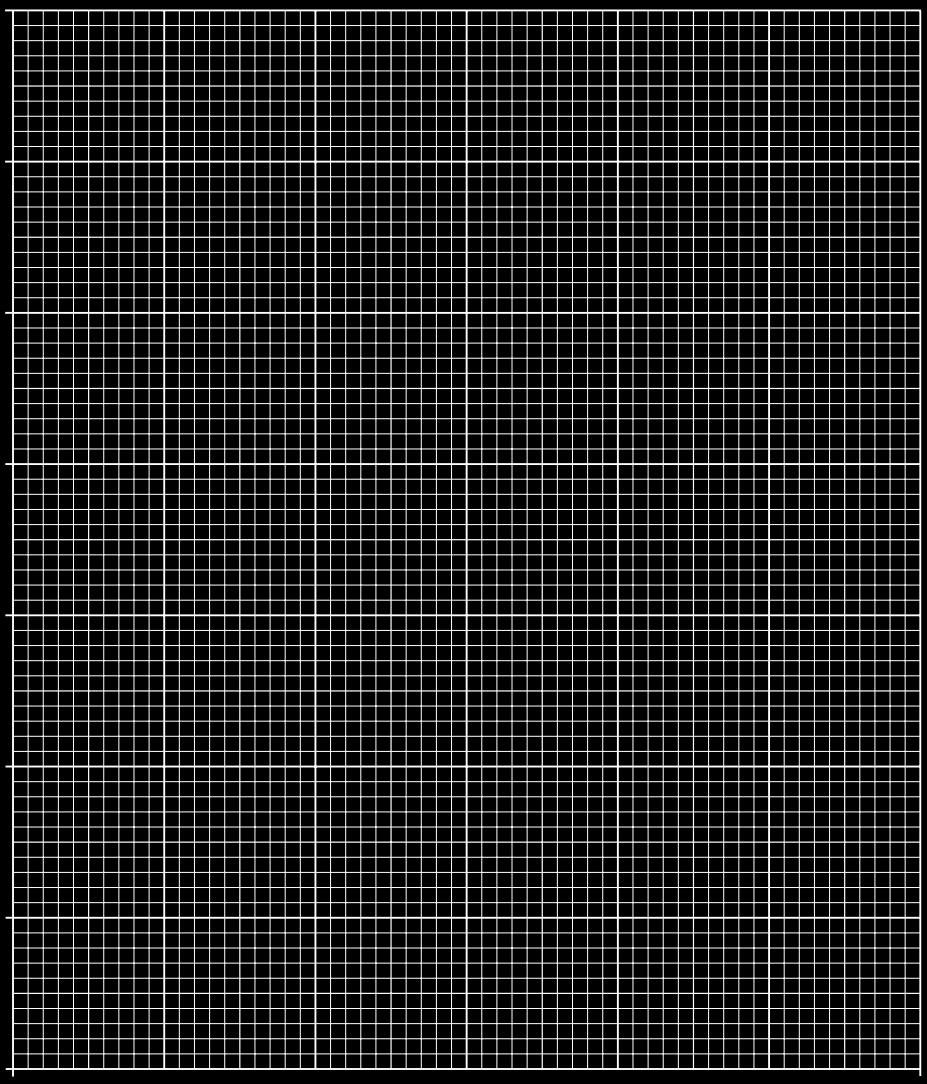
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They collected the following data.

Rain collected (litres)

Day	Site A	Site B	Site C	Site D
Day 1	19	17	14	18
Day 2	31	29	25	30
Day 3	10	8	6	9
Average	20	18	15	19

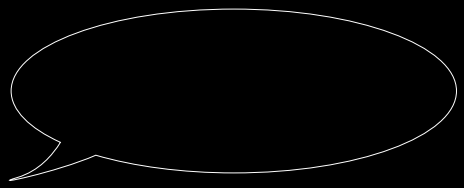
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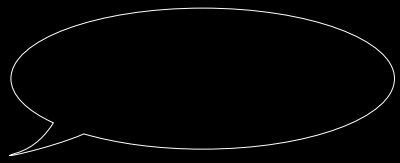


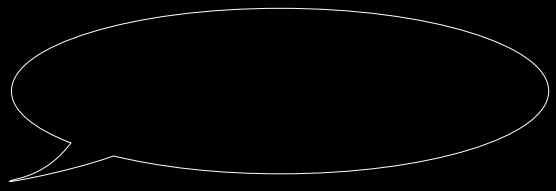
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Fig. 10

Coconuts are an important export crop and 90% of Sri Lanka's production comes from small farms about 2 hectare in size. The coconut trees are planted at 140 trees per hectare. Other plants can be grown between the trees and livestock are tied to the trees (Fig. 10). The annual income for one small farm is shown in Fig. 11.

Farm produce	Annual income Rs
vegetables	300
coconuts	3400
pasture plants (peas, beans, some grasses)	1800
cattle	31500
chickens	8000
total	45000

Fig. 11

(a) (i) What percentage of total annual income comes from cattle?

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